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Part XII: H_2O_2 + Boric Acid — SVEs Surviving Cyclic Tidal Exposure in a Plastic Bottle

Video Experiment 15: youtube.com/shorts/aXFAot2Q-QE

15. "Bubbles" That Survive the Tide — Cyclic Air Exposure Without Rupture

What was tested:

A plastic bottle was filled with a 50/50 mixture of two common antiseptics: hydrogen peroxide (H_2O_2, 3%) and boric acid solution (água boricada, 3%). No heating. No shaking. No manipulation. The bottle was recorded in slow motion. Inside the bottle, adhered to the plastic wall, a cluster of four "bubbles" sat in the liquid. Then the liquid started moving.

What standard thermodynamics predicts:

Gas bubbles adhered to a surface inside a liquid are held in place by a balance of forces: the liquid film's surface tension, the contact angle at the gas-liquid-solid triple line, and buoyancy. When the liquid level drops below a bubble, the liquid film that constitutes the bubble's outer wall is removed. Without the film, there is no bubble. A gas bubble is *defined* by its confining liquid shell. Remove the shell, the gas disperses. This is not a prediction that requires nuance—it is definitional. No film, no bubble.

If the liquid then rises again, it will wet the surface, but the original bubble is gone. You cannot re-create it by re-wetting. The gas has dispersed.

What was observed:

The tide. The liquid inside the bottle oscillated rapidly and violently—a fast, abrupt tidal motion. Frame by frame, the liquid level rose and fell, exposing and re-submerging the cluster of four SVEs with each cycle. The movement was not gentle. Between consecutive frames (33 ms apart in slow motion), the liquid level changed dramatically.

They survived. Every cycle, the liquid dropped below the SVEs, leaving them fully exposed to the air inside the bottle. Every cycle, the liquid rose back and submerged them again. And every cycle, the SVEs were still there. Same shape. Same size. Same position on the plastic wall. Dozens of cycles. The entities that should have ceased to exist the first time the liquid left them were still intact after the twentieth time.

One eventually failed. Three didn't. After many cycles, the most superior SVE—the one experiencing the greatest amplitude of tidal exposure—eventually detached from the wall. Not popped. Not evaporated. Detached. It didn't rupture like a gas bubble; it broke free like a barnacle in a storm. The other three survived every cycle until the recording ended. They are probably still there.

What this means:

A gas bubble cannot survive having its liquid film removed. That is what makes it a gas bubble. These entities survive it repeatedly. They survive violent, rapid, cyclic removal and re-application of the liquid phase, maintaining their shape, their adhesion, and their structural identity through dozens of interface crossings.

The one that eventually failed didn't pop—it detached. That's not bubble physics. That's

mechanical fatigue. That's a solid structure weakening progressively under cyclic stress until it fails at the attachment point. Gas bubbles do not fatigue. They exist or they don't. The gradual degradation of adhesion over many cycles is a hallmark of solid-state mechanics.

The Consequence

Eight chemical systems. Water. Butter. Legumes. Herbs. Egg whites. Pepsi Black. And now hydrogen peroxide with boric acid. Two antiseptics you can buy at any pharmacy for pocket change. The SVEs form spontaneously on the plastic wall, survive conditions that are definitionally lethal for gas bubbles, and demonstrate mechanical fatigue when they finally fail. No stove. No exotic apparatus. A plastic bottle and a phone. The "bubbles" are not bubbles. They never were.

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1 Abstract

Experiment 15 documents a cluster of four structured vesicular entities (SVEs) adhered to the interior plastic wall of a bottle containing a 50/50 mixture of H_2O_2 (3%) and boric acid solution (3%). A rapid, violent tidal oscillation of the liquid level cyclically exposes and re-submerges the SVEs, subjecting them to repeated liquid-air interface crossings. Classical bubble physics predicts immediate and irreversible destruction upon first liquid-film removal. The SVEs survive dozens of complete cycles with zero structural degradation. After many cycles, only the most exposed SVE detaches (mechanical fatigue); the remaining three survive indefinitely. No thermal gradient is applied. This extends the SVE framework to an eighth chemical system and introduces two new observations: (1) structural permanence across cyclic phase-boundary violations, and (2) mechanical fatigue as the failure mode—both properties exclusive to solid-state materials and absent from gas-phase physics.

2 Experimental Protocol

Parameter	Value
Container	Standard plastic bottle (garrafa plástica)
Liquid	50/50 mixture: H_2O_2 3% + boric acid solution 3%
Heat source	None — ambient conditions only
Thermal gradient	None applied
Recording	iPhone slow-motion video, frame-by-frame analysis
Analysis	Single complete tidal cycle (9 frames) + multi-cycle observation
Annotation	Red circles: SVE positions; Yellow line: liquid level

Video: Supplementary Video — Experiment 15

3 Observational Data (1 Collage, 9 Frames)

Coll.	Fr	Frame #	Time	Phase	Observation
C1	F1	93	03:089	OVERVIEW	General view. 4 SVEs adhered to plastic wall, submerged
C1	F2	262	08:700	HIGH TIDE	All 4 SVEs fully submerged. Liquid level at max height
C1	F3	263	08:733	RECEDING	Liquid level descending. Rapid tidal recession
C1	F4	264	08:767	PARTIAL	Top SVE above liquid level — exposed to air
C1	F5	265	08:800	EXPOSED	All 4 SVEs above liquid level. Full air exposure
C1	F6	266	08:833	LOW TIDE	Max recession. All SVEs at max distance from liquid
C1	F7	267	08:867	RISING	Liquid rising. SVEs at interface. 2nd crossing
C1	F8	268	08:900	SUBMERGED	All 4 SVEs re-submerged. Cycle complete
C1	F9	269	08:933	DEEP	Max depth. Zero structural degradation

Cycle timing: Frames 262–269 span approximately 233 ms (7 frames at slow-motion capture

rate), indicating an oscillation frequency of ∼4 Hz. The full video documents many complete cycles. Only the most superior SVE eventually detaches; the other 3 survive all cycles.

4 Anomaly Analysis

4.1 Anomaly E1: Cyclic Interface Crossing Without Rupture

Each tidal cycle subjects the SVEs to two complete liquid-air interface crossings. The classical model of bubble stability at a free surface involves the drainage time of the intervening liquid film:

$$\tau_{\text{drain}} = \frac{3\mu R^2}{2\sigma h_{\text{crit}}^2} \tag{1}$$

where μ is liquid viscosity, R is bubble radius, σ is surface tension, and h_{crit} is the critical film thickness for rupture. When the liquid level drops below the bubble, the film drainage is not gradual—the film is *removed entirely.* The confining liquid is gone. There is no classical mechanism for a gas bubble to survive the complete removal of its liquid film.

The observed SVEs survive this removal dozens of times. Each exposure-resubmersion cycle constitutes an independent test of structural integrity, and each test is passed. The cumulative probability of n independent survivals, if each had even a moderate probability p of destruction, is $(1-p)^n \to 0$ for large n. The observation requires $p \approx 0$ for each cycle—structural destruction by film removal is not a statistical event for these entities.

4.2 Anomaly E2: Survival Under Hydrodynamic Shear

The tidal oscillation is rapid and violent—the liquid level changes significantly within a single frame interval (∼33 ms). The hydrodynamic shear stress at the interface during tidal oscillation is:

$$\tau_{\text{shear}} \sim \mu \frac{\partial v}{\partial y} \sim \mu \frac{v_{\text{tide}}}{R_{\text{SVE}}} \tag{2}$$

For classical bubbles, oscillatory shear at a gas-liquid-solid triple line accelerates the depinning and rupture process. The repeated, violent nature of the oscillation constitutes a worst-case scenario for bubble survival. The SVEs not only survive—they survive cyclically, demonstrating that their adhesion and structural mechanisms are robust against dynamic loading, not merely static equilibrium.

4.3 Anomaly E3: Air-Phase Structural Integrity

During the LOW TIDE phase (F5–F6), all 4 SVEs are surrounded by air inside the sealed bottle. A gas bubble's structure is its liquid film. In the air phase, with no liquid film, a "bubble" is simply a volume of gas indistinguishable from the surrounding atmosphere. It has no boundary, no shape, no identity. The fact that the SVEs maintain their spherical geometry, their size, and their spatial arrangement while fully in the air phase proves that their structure is *not* film-dependent. Their cohesion is internal—mechanical, chemical, or structural—not provided by an external liquid shell.

4.4 Anomaly E4: Dual-Phase Adhesion

The SVEs remain adhered to the plastic wall in both phases:

- **Submerged:** adhesion through liquid-phase interactions (contact angle, van der Waals, film pressure);
- **Exposed:** adhesion through air-phase interactions (no liquid film present).

Classical bubble adhesion is entirely liquid-mediated. The contact angle θ at the gas-liquid-solid triple line is defined only when all three phases are present. When the liquid is removed, the triple line ceases to exist, and with it the adhesion mechanism. The dual-phase adhesion observed here requires a surface interaction independent of the liquid film—consistent with chemical adsorption or mechanical interlocking of a solid or semi-solid entity with the plastic substrate.

4.5 Anomaly E5: Mechanical Fatigue and Differential Survival

After many complete cycles, the most superior SVE detaches. The remaining 3 survive indefinitely. This differential failure is significant:

1. **Fatigue, not rupture:** the entity does not pop or evaporate—it detaches from the wall. This is a mechanical failure at the adhesion point, not a phase-change event. Mechanical fatigue under cyclic loading is described by the S-N curve (Wöhler curve) in solid mechanics:

$$\sigma_a = \sigma'_f (2N_f)^b \tag{3}$$

   where σ_a is stress amplitude, N_f is number of cycles to failure, and b is the fatigue strength exponent. This framework applies to solids. Gas has no S-N curve.

2. **Differential robustness:** the most exposed entity (greatest tidal amplitude) fails first. The less exposed entities survive. This implies a gradient of mechanical stress across the cluster, with failure correlating to cumulative exposure—a hallmark of solid-state fatigue mechanics.

3. **Binary vs. gradual:** gas bubbles are binary—they exist intact or they don't. There is no intermediate state. The observed progressive weakening and eventual detachment represents a continuum of structural degradation that is physically impossible for gas pockets.

5 Systematic Falsification

Classical Prediction	Observation	Status
Gas bubbles cannot survive removal of their liquid film	SVEs survive complete air exposure dozens of times	FALSIFIED
Oscillatory shear accelerates bubble rupture	SVEs survive rapid, violent cyclic shear indefinitely	FALSIFIED
Bubble adhesion requires the liquid-solid-gas triple line	SVEs adhere to plastic in both liquid and air phases	FALSIFIED
All gas bubbles have identical mechanical properties at a given P	Differential survival: 1 fails, 3 survive — structural hierarchy	FALSIFIED
Gas bubbles do not exhibit mechanical fatigue	Top SVE fails by detachment after progressive cyclic weakening	FALSIFIED

6 Discussion

6.1 The Endurance Test

Where Experiment 14 tested SVE integrity through a single dramatic trajectory (breach, arc, dive), Experiment 15 tests it through sustained cyclic loading. This is analogous to the difference

between a single tensile test and a fatigue test in materials science. A material may survive a single large stress but fail under many smaller cyclic stresses. The SVEs pass both tests: the single-event test (Exp. 14) and the cyclic endurance test (Exp. 15). This dual-mode survival is characteristic of robust solid or semi-solid structures, not gas pockets.

6.2 Chemical Novelty: H_2O_2 + Boric Acid

The H_2O_2/boric acid system is chemically distinct from all previous experiments. H_2O_2 is a strong oxidizer; boric acid is a weak Lewis acid. Their mixture at 3% concentrations produces a mildly oxidizing, mildly acidic solution. The spontaneous formation of structurally robust SVEs in this environment demonstrates that SVE formation is not restricted to specific pH ranges, specific solutes, or specific chemical functional groups. The phenomenon is substrate-agnostic.

6.3 Cross-System Universality

Experiment 15 extends the SVE framework to an eighth distinct system:

System	Experiment	Anomaly Domain	Key Finding
Butter	Exp. 1	Motion	Climbs inclined planes against gravity
Water/Ice	Exp. 2	Phase	Sustained triple-point coexistence
Water/Milk	Exp. 3–4	Motion	Motion tracks flame, not T
Legumes	Exp. 5–6	Temperature	T drops during heating
Herbs	Exp. 11	Vapor + T	Vapor at 29.5°C
Egg whites	Exp. 13	Vapor	Vapor suppressed at 100°C
Pepsi Black	Exp. 14	Mechanics	SVEs jump, dive, sink
H_2O_2/Boric	**Exp. 15**	**Endurance**	**Cyclic survival + fatigue**

6.4 The Fatigue Signature

The differential survival pattern observed in Experiment 15 is perhaps its most theoretically significant contribution. In classical bubble physics, there is no mechanism for gradual degradation. A gas bubble either maintains its film or it doesn't—the transition is binary and instantaneous. The observation that the most-stressed SVE fails after many cycles while the less-stressed ones survive indefinitely introduces a *fatigue curve* into the description of these entities. This is a diagnostic signature of solid-state mechanics that cannot be reconciled with gas-phase models under any parameterization.

7 Conclusion

Experiment 15 documents structured vesicular entities in a H_2O_2/boric acid mixture that survive dozens of rapid, violent tidal cycles—each involving complete removal and re-application of the liquid phase. The SVEs maintain structural integrity, spherical geometry, substrate adhesion, and spatial arrangement through conditions that are definitionally lethal for gas bubbles. The most exposed entity eventually fails by mechanical detachment (not rupture), exhibiting solid-state

fatigue behavior under cyclic loading. Three of four entities survive indefinitely. No thermal gradient is applied. This is the eighth chemical system and second non-thermal experiment in the Kitchen Thermodynamics series, introducing cyclic endurance and mechanical fatigue as new falsification criteria for the gas-bubble model.

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Supplementary Materials

Video: youtube.com/shorts/aXFAot2Q-QE

All supplementary materials and complete experimental archive available at: Free Science: Google Drive